



Kointoss Solutions

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Strategy

1. Perform regression on the dissociation constants
2. Use a Neural Network that receives a protein and a molecule pair as input to predict their interaction
3. From the Kd extract the selectivity scores

Workflow



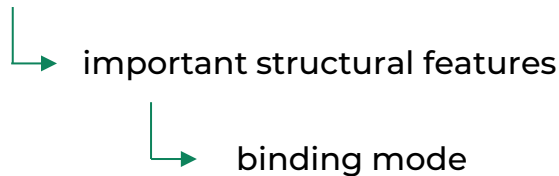
- Ensure data is usable, remove bias.

- Select valuable features, reduce to most relevant.

- Build neural network

Data Quality Control

- **Inhibitor-kinase binding data** (K_d) was left untouched.
 - No outliers or duplicates, blank values already substituted
- We considered annotating **undetermined** inhibitor **binding types**.
 - Molecular fingerprint data



Feature Selection – kinases



Mutation status:

Inhibitor binding site
mutations



Kinase group

Conserved sequence
(homology)



Evolutionary family



ProteinBert

Amino acid sequence
embeddings – describes
kinase structure

Feature Selection – inhibitors

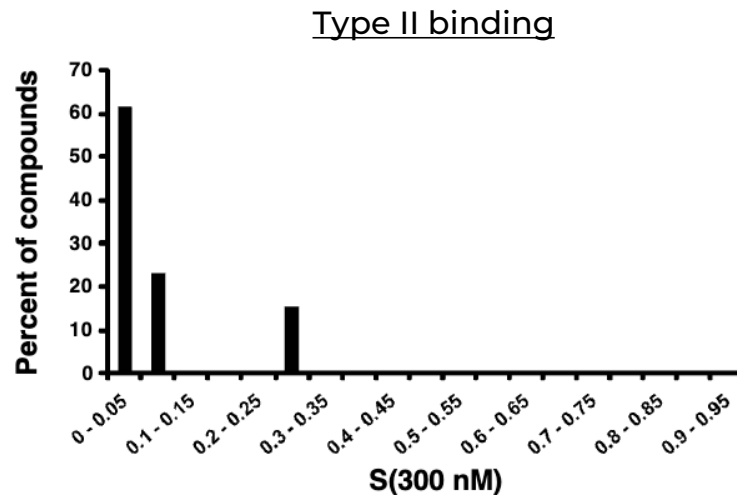
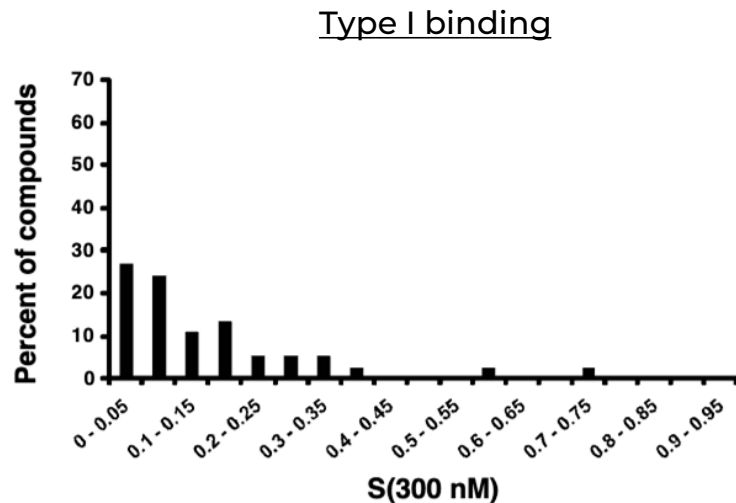
Factors that would **impact binding** & are **selectivity for a given inhibitor/protein**

Use RDKit to extract inhibitor features:

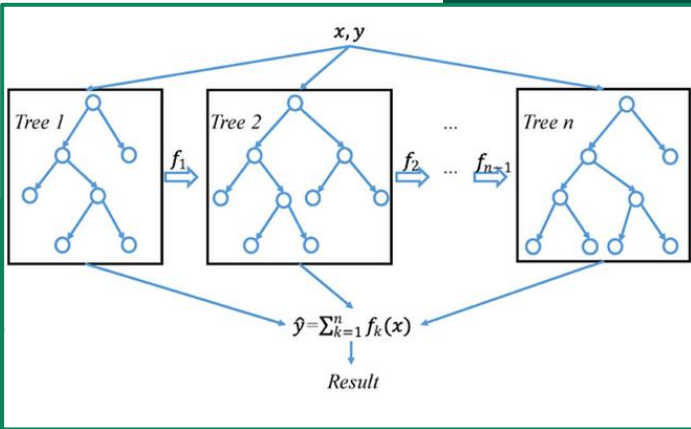
- LogP (lipophilicity)
- Molecular Weight
- Rotatable bonds, Hydrogen bond donors/acceptors
- Topological polar surface area
- Number + size of aromatic ring systems
- Chirality information
- Bertz molecular complexity index
- **Morgan Fingerprint** (ECFP4) -> 2048 bit string

Feature Selection – inhibitors

Binding type: Type I, Type II, Undetermined



Binding type
matters



Our strategy

is to gather **as many relevant features** as possible, and then filter for the most relevant features using **XGBoost**.

Dimensionality Reduction

PROBLEM: Our number of features had grown to **17672**

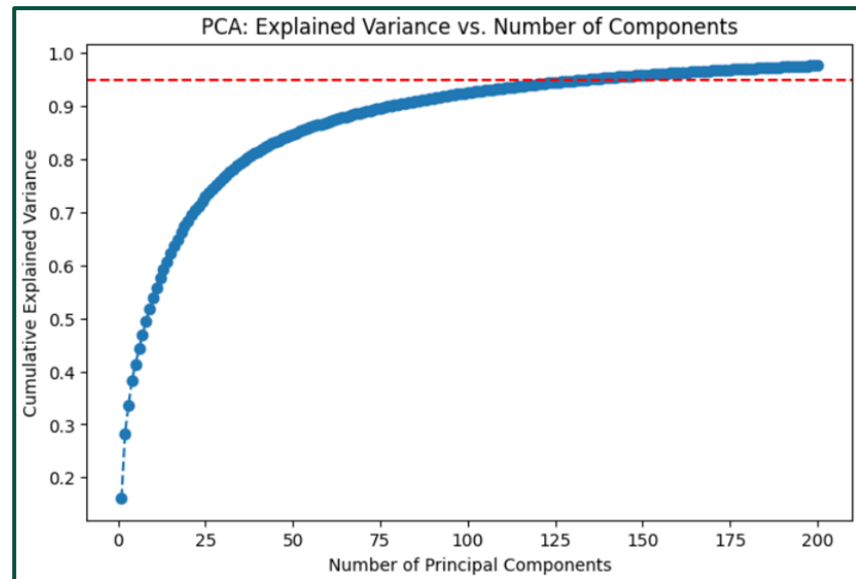
Use PCA on the protein embeddings

- Show which features are the most important
- Result: Kept the top 150

Train the ECFP4 data on a XGBoost model:

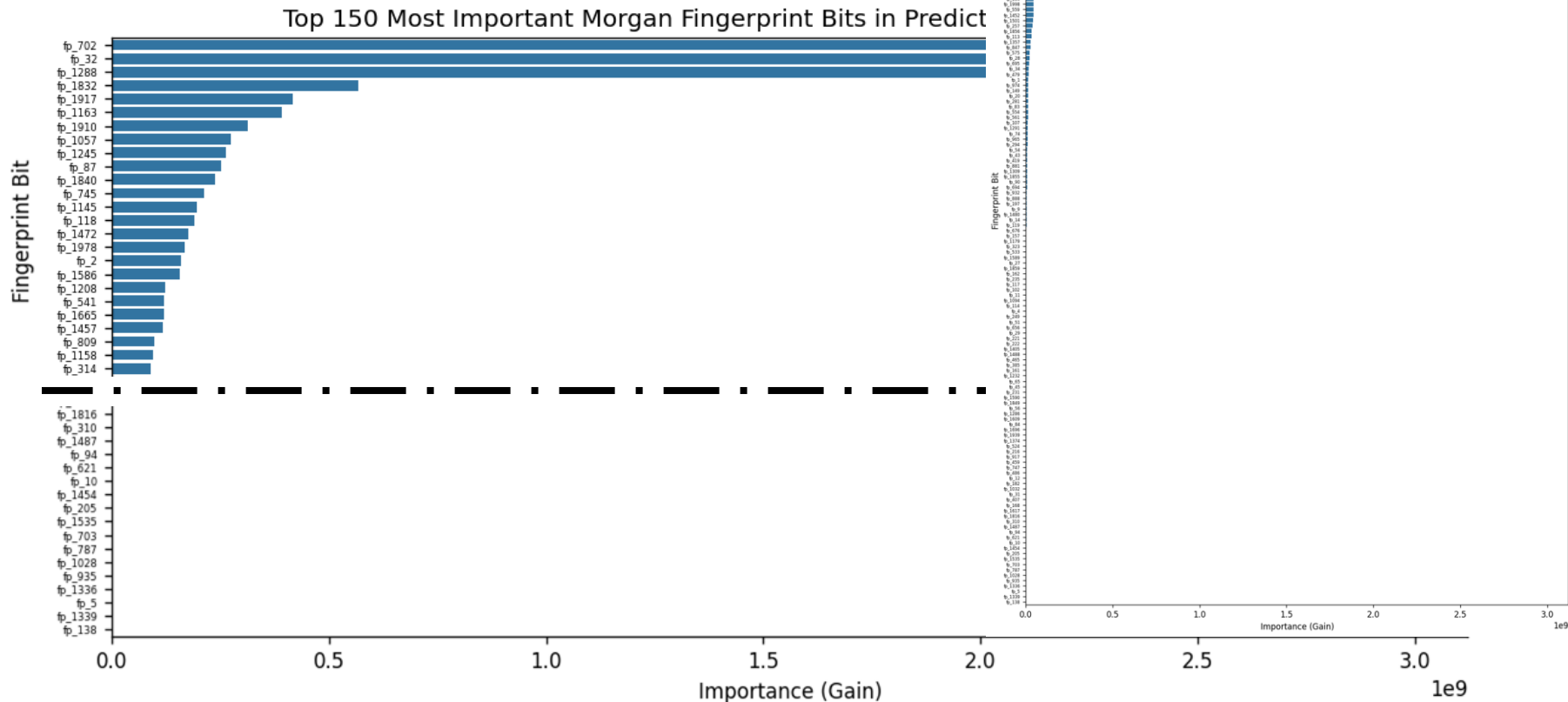
- Show which bits yield the most impact
- Result: Kept the top 50

RESULT: Reduced the labels to a total of **224**



Dimensionality Reduction

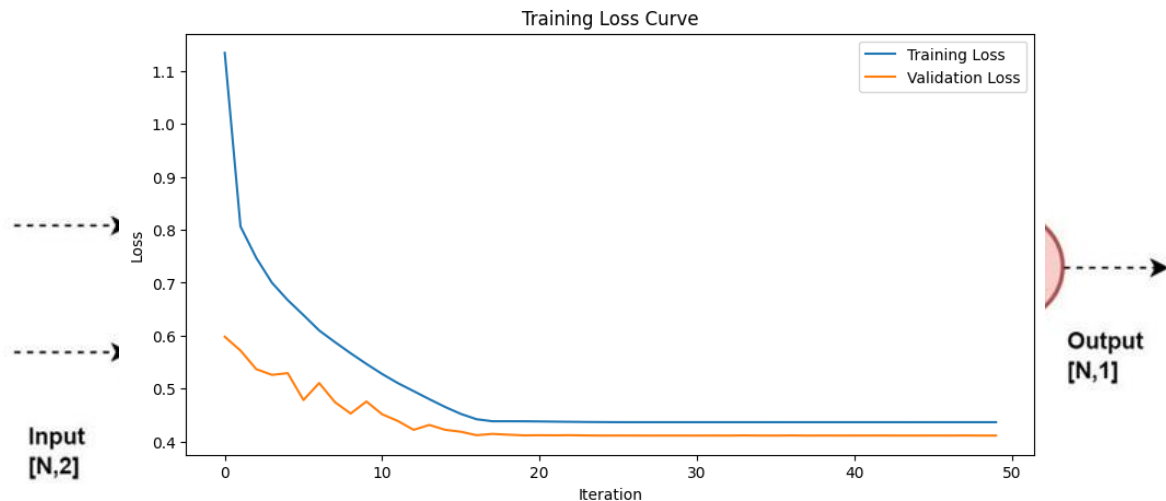
PROBLEM: Our number of features had grown to **17672**



Base Learning Model

Decided on using a neural network:

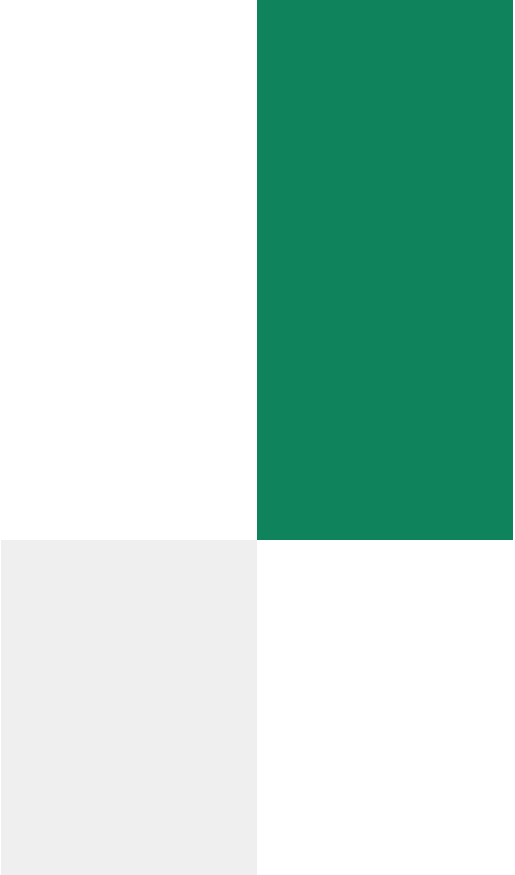
- Three hidden layers with 512, 512, 64 nodes
- ReLU activation function
- Mini-batch gradient descent + ADAM
- Batch Normalization & Dropout





73.6%

classification accuracy on testing data



Improving The Model

What we were working on
before running out of time

Future Directions

Features

Kinase features

- Use **local** embeddings
- Omit explicit **kinase group** labels

Inhibitor features

- Improve **binding mode** annotations
- Optimize **fingerprint** use

Model

Graph-based model

- Increased **fidelity**
- Would have taken **too long** to train

Fine-tuning

- Varying the **number** of hidden layers

Kointoss Solutions

presents:

kiNNet

A biochemistry **informed neural network** for kinome
selectivity prediction of small molecule inhibitors

(patent pending)

Citations

- <https://medium.com/@thommaskevin/tinyml-xgboost-regression-d2b513a0d237>
- Davis, M., Hunt, J., Herrgard, S. *et al.* Comprehensive analysis of kinase inhibitor selectivity. *Nat Biotechnol* 29, 1046–1051 (2011)

